Product Requirements Document

Coba Raket

Try tennis rackets before you buy, with a verifiable history for every racket.

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Status	Ready for demand testing. Not yet validated with customers.
First market	Yogyakarta, Indonesia

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1. Executive summary

Coba Raket lets tennis players in Yogyakarta try up to three rackets for three days before they buy one. An AI matcher shortlists rackets that are in stock, a short QR handover records each racket's condition, and every racket carries a public digital passport of its trials, repairs and inspections.

Buying a racket today is an expensive guess. A Wilson beginner model costs about Rp784,000, while a Babolat Pure Drive Team costs about Rp3,950,000 [1][2]. Overseas retailers solve this with paid demo programs, and some credit the fee toward a purchase [3][5]. This research did not find an equivalent program in Indonesia, although the search was not exhaustive.

Item	Summary
Customer	New players buying a first proper racket, and regular players upgrading
Promise	Find the right racket by playing with it, not by guessing
Core offer	Three-day trial, refundable deposit, trial fee credited if the customer buys
Differentiators	Local stock, AI matching limited to real inventory, coach partners, verifiable racket history
Revenue	Trial fees, used racket sales, new racket referrals, stringing and accessories
Start	One-week manual demand test with 3 to 4 borrowed rackets, before any code
Product build	Two-week mobile web MVP, with the passport running on a public testnet
Starting cost	About Rp600,000 on borrowed rackets, or about Rp7,800,000 with 6 owned rackets (estimates)
Main risk	Beginners may prefer to buy a cheap racket rather than pay for a trial

Decision requested

Approve the one-week demand test in Section 15. Build only if it reaches at least 10 paid racket-trials, 3 customers seriously ready to buy, and 1 coach or court partner.

How to read this document

Facts from outside sources carry a number in square brackets, matched to the list in Appendix D. Prices, targets and policies written as **proposed** or **assumed** are working hypotheses for the pilot, not findings. A **racket-trial** means one racket on one trial, so a customer testing three rackets creates three racket-trials.

2. Problem and opportunity

2.1 Problem statement

Players must commit Rp800,000 to Rp4,000,000 to a racket before knowing how it feels in real play. Weight, balance, head size and grip affect comfort and control, but specification lists mean little to a new player. The result is a wrong purchase, a racket that stays unused, or a resale at a loss.

Used rackets are cheaper but carry a second problem. Online buyers rarely can test a racket first, and they rely on the seller's word about cracks and past repairs [13]. Convincing counterfeits also exist [14].

2.2 Evidence so far

Claim	Evidence	Strength
The price of a wrong choice is meaningful	Listings show about Rp784,000 for a Wilson beginner racket and Rp3,950,000 for a Babolat Pure Drive Team [1][2]	Medium: listing prices change
People pay to try rackets	Tennis Warehouse charges $25 for up to three racquets and credits it on purchase. RacquetGuys charges $10 per racquet [3][5]	Strong overseas, unproven in Indonesia
Trials can be run safely	Overseas programs use deposits, ID checks and card holds [4][5]	Strong overseas
Yogyakarta has an active scene	Several courts charge about Rp30,000 to Rp60,000 per session [8]. A 2023 event article said court schedules were often full [9]	Medium: the article is dated and promotional
Young players are entering tennis	A 2026 article describes tennis as a social and networking activity for young people in Jakarta [10]	Weak for Yogyakarta
A used racket market exists	Preloved rackets are listed at about Rp1,900,000 to Rp3,500,000, with cosmetic grades on some listings [11][12]	Medium: Carousell seller data is a small sample
The local gap is open	No Indonesian demo program was found in this research	Weak: absence of evidence only

2.3 What is still unknown

- Whether Yogyakarta players will pay a trial fee when they could borrow a friend's racket.
- How many people buy a racket in this city each month.
- Whether courts and coaches already lend rackets for free. At least one venue, Hulahoop Wellness, offers racket rental [8].
- Whether a deposit of Rp250,000 or more stops students from booking.
- Whether used racket buyers trust, or even want, a digital passport.

2.4 Existing alternatives

Alternative	Strength	Weakness	Coba Raket response
Buy new from a store or marketplace	Simple, warranty, wide choice	No real play test	Trial first, then buy through a partner
Borrow from a friend or coach	Free and trusted	Limited models, no shortlist	Structured comparison of up to 3 rackets
Rental at a court	Available at the point of play	Designed for a session, not a purchase decision. Local selection not confirmed	Multi-day trial, advice and fee credit
Preloved sellers	Lower price	Usually no test, and history depends on the seller	Used stock with inspection history
Overseas demo programs	Proven model	Aimed at US and EU customers	Local stock and pickup
Racket rental software such as RentRacket	Ready for clubs [15][16]	A tool for venues, not a consumer service	Could become a competitor or a future supplier

2.5 Positioning

For tennis players about to spend real money on a racket, **Coba Raket** is a local try-before-you-buy service **that** matches them to rackets and lets them play with those rackets for three days. **Unlike** marketplaces and preloved sellers, it replaces guesswork with real play and a checkable racket history.

3. Goals and non-goals

3.1 Customer goals

- Help a player choose a racket with confidence within one trial.
- Make costs, deposit and damage rules clear before payment.
- Give used buyers a history they can check themselves.

3.2 Business goals

- Prove that people will pay for trials in one city before buying more stock.
- Reach about 24 racket-trials a month from 6 rackets by the second pilot month.
- Lose no more than the value of one racket during the pilot.

3.3 Portfolio goals

The MVP should demonstrate full-stack web development, a useful and constrained AI agent, and a practical Web3 feature. It should be live, used by real people, and documented with usage numbers and failed attack demos.

3.4 Non-goals for the MVP

- Hourly on-court rental, delivery, or self-service lockers.
- Padel, badminton, pickleball and junior rackets.
- A marketplace for rackets owned by other people.
- Certifying that a racket is genuine. The passport records history, not authenticity.
- Crypto payments, tokens, NFTs, wallets or smart contract escrow. Customers pay in rupiah and never touch crypto.
- Medical advice about arm pain or injury, and automatic damage decisions by AI.

4. Users and needs

User	Situation	Main need	Success looks like
First racket buyer	Student or young worker who has played with borrowed rackets. Budget assumed at Rp700,000 to Rp2,000,000	Avoid wasting money on the wrong racket	Knows which racket to buy after one trial
Upgrader	Plays weekly and is considering a Rp2,000,000 to Rp4,000,000 racket	Compare options in real matches	Confidently chooses or avoids an expensive upgrade
Coach	Is asked what students should buy	A simple way to let students test rackets	Refers students and earns a fee
Used racket buyer	Wants a lower price without hidden damage	Trustworthy condition history	Buys after checking the passport
Operator	Runs stock, handovers and refunds, initially the founder	Fast, consistent process with few disputes	Handover under 60 seconds per racket, clear records

Key user stories

- As a beginner, I want a short list of rackets that suit me, so I do not need to understand every specification.
- As a customer, I want to see the fee, deposit and damage rules before paying, so nothing surprises me.
- As a customer, I want proof of the racket's condition at pickup, so I am not blamed for existing damage.
- As an operator, I want to record a handover in under a minute, so the process works on a busy court.
- As a used buyer, I want to verify that a racket's history was not changed later.
- As a coach, I want a referral link, so I can recommend trials and track what I earned.

5. Product concept

The service has four parts: **Match**, **Trial**, **Decide** and **Passport**. The core value comes from the first three. The passport supports trust and resale, and the trial must still work if it is temporarily down.

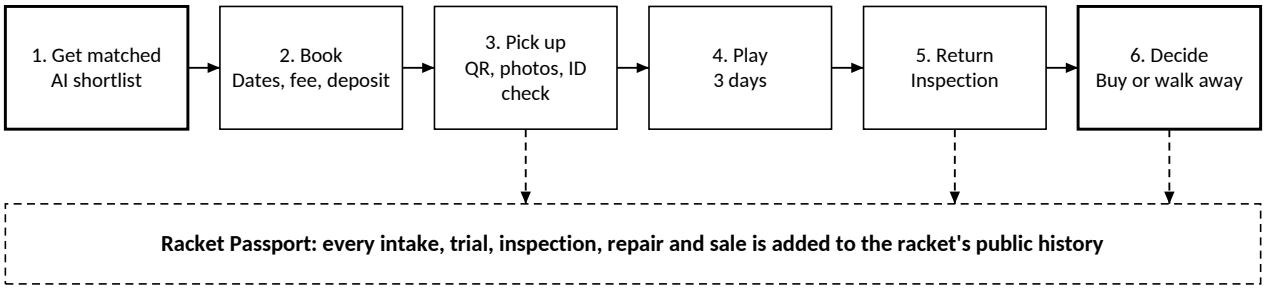


Figure 1. Customer journey and passport events.

5.1 Happy path

1. Dina answers six questions and receives three in-stock rackets with plain explanations.
2. She chooses two rackets, selects a pickup slot, verifies her phone and accepts the rental terms.
3. She pays Rp100,000 in trial fees and a Rp500,000 refundable deposit.
4. At the court, staff scan each racket, photograph it, check her ID in person and record the condition. She confirms on her phone.
5. She plays for three days and returns both rackets. Staff inspect them and record normal wear.

6. The app refunds her deposit within 24 hours, and offers the fee as a discount if she buys within 14 days.
7. Each racket's passport gains a new event. No personal data is included.

5.2 Scope by phase

Capability	Phase 0: Demand test	Phase 1: MVP	Phase 2: After pilot
Stock	3 to 4 borrowed rackets	6 to 8 owned or partner rackets	Multiple grip sizes and partner stores
Booking	Form and WhatsApp	Web booking and calendar	Coach and court booking links
Matching	Founder or coach advice	AI matcher on real inventory	Learns from trial outcomes
Payment	Manual QRIS, manual refund	Manual QRIS with ledger. Gateway if ready	Licensed gateway with automatic refunds
Handover	Paper checklist and phone photos	QR handover in the operator console	Partner staff accounts at several courts
Passport	Not included	Testnet attestations and public verifier	Mainnet only if buyers value it
Purchase	Referral to a store	Fee credit and used stock sales	Partner commissions and direct new stock

6. Functional requirements

Priority uses **Must** for MVP launch, **Should** for MVP if time allows, and **Could** for later.

6.1 Catalogue

ID	Requirement	Priority	Acceptance criteria
CAT-1	List rackets with photo, model, weight, head size, grip, condition grade, trial fee, deposit and availability.	Must	Every value comes from the database. Unavailable rackets show the next free date.
CAT-2	Filter by level, weight, grip size and purchase budget.	Should	Filters combine and return results in under 1 second on 4G.
CAT-3	Show a racket detail page with specifications and a passport link.	Must	The page opens from the catalogue and from the racket's QR code.

6.2 AI matcher

ID	Requirement	Priority	Acceptance criteria
AI-1	Ask up to six guided questions, with optional free text.	Must	A new user finishes in under 2 minutes.
AI-2	Return one to three rackets that are in stock for the chosen dates.	Must	No result is retired, reserved or outside the user's limits.
AI-3	Explain each choice in simple Bahasa Indonesia using stored specifications only.	Must	Each number shown matches the database. Automated test passes.
AI-4	Fall back to rules if the model fails.	Must	With the LLM disabled, users still get a shortlist.
AI-5	Handle arm or injury mentions safely, without diagnosis.	Must	Reply suggests a coach or health professional and does not store the detail.
AI-6	Ignore instructions in free text that try to change prices, stock or rules.	Must	Prompt injection test set passes.

6.3 Booking

ID	Requirement	Priority	Acceptance criteria
BK-1	Select up to three rackets and one pickup and return slot.	Must	New customers cannot select Tier 3 rackets.
BK-2	Prevent double booking and keep one turnaround day between trials.	Must	A database constraint rejects overlapping bookings, including under concurrent requests.
BK-3	Verify the customer's phone by one-time code.	Must	Unverified users cannot confirm a booking.
BK-4	Record acceptance of a specific version of the rental terms.	Must	Stored with timestamp, version and booking ID.
BK-5	Let customers cancel or reschedule under the policy in Section 7.	Must	Fee and refund outcome are shown before confirmation.
BK-6	Let customers request an extension.	Could	Allowed only when the racket is free.
BK-7	Show live booking status.	Must	States match Figure 5.

6.4 Payments and deposits

ID	Requirement	Priority	Acceptance criteria
PAY-1	Show trial fee, deposit and refund timing separately before payment.	Must	Displayed on the booking summary.
PAY-2	Accept QRIS, confirmed manually by the operator in the MVP.	Must	Booking moves to Confirmed only after operator confirmation.
PAY-3	Use a licensed gateway with webhooks.	Could	Duplicate webhooks do not create duplicate payments.
PAY-4	Keep an append-only ledger for fees, deposits, refunds, charges and credits.	Must	Each balance can be rebuilt from ledger rows.
PAY-5	Track deposit refunds.	Must	Refund due and paid times are stored. Target is within 24 hours of inspection.
PAY-6	Never store full card, bank login or wallet credentials.	Must	Security review passes.

6.5 Handover and return

ID	Requirement	Priority	Acceptance criteria
HO-1	Scan the racket QR and match it to the booking.	Must	A wrong racket shows a clear error and cannot proceed.
HO-2	Capture front and back photos from the camera.	Must	Gallery upload is disabled. Time and operator are recorded.
HO-3	Complete a checklist for frame, throat, grommets, strings and grip.	Must	All items are required.
HO-4	Record that ID was checked, without storing the ID number or image.	Must	Database stores a true or false flag only.
HO-5	Both sides confirm the handover.	Must	Customer confirms from their verified phone. Without it, the booking cannot become Active.
HO-6	Record a new overgrip or cleaning at each turnaround.	Should	Racket cannot be marked ready until complete.
HO-7	Support return with the same flow.	Must	Median time is under 60 seconds per racket in testing.

6.6 Damage and disputes

ID	Requirement	Priority	Acceptance criteria
DMG-1	Show pickup and return photos side by side.	Must	Operator sees both before choosing a grade.
DMG-2	Require evidence for any charge.	Must	A charge cannot be saved without at least one return photo and a reason.
DMG-3	Give the customer 48 hours to dispute.	Must	Customer sees evidence and can respond. Deposit remainder is refunded meanwhile.
DMG-4	Use a reviewer for disputes.	Should	A coach or partner not involved in the booking records a decision.
DMG-5	Do not charge if pickup photos cannot clearly show the area.	Must	Rule appears in the operator flow.
DMG-6	Retire cracked or bent rackets from trials.	Must	Status changes to Retired and the racket disappears from the catalogue.

6.7 Racket passport

ID	Requirement	Priority	Acceptance criteria
PP-1	Provide a public page for each racket.	Must	Opens without login and shows no personal data.
PP-2	Record intake, trial, inspection, damage, restring, repair, sale and retirement events.	Must	Each event has a sequence number and previous hash.
PP-3	Anchor each event as an EAS attestation on Base Sepolia.	Must	Attestation ID appears on the event.
PP-4	Verify the history from the browser.	Must	Editing any stored event shows a mismatch.
PP-5	Detect missing events.	Must	A sequence gap shows a clear warning.
PP-6	Do not block handover if the chain is unavailable.	Must	Events queue, retry and show Pending.
PP-7	Give customers a receipt hash after each trial.	Should	Customer can later check that their event still exists.

6.8 Decision and purchase

ID	Requirement	Priority	Acceptance criteria
BUY-1	After return, ask the customer to rate each racket and decide.	Must	Options are buy new, buy used, or not now.
BUY-2	Credit the trial fee if they buy within 14 days.	Must	Credit appears once in the ledger.
BUY-3	Track partner referrals.	Should	Each sale records source and commission.
BUY-4	Sell retired trial rackets with their history.	Should	Sale adds an event and closes the racket for trials.

6.9 Admin and notifications

ID	Requirement	Priority	Acceptance criteria
ADM-1	Manage racket inventory and status.	Must	Status options are Ready, Reserved, Out, Servicing, Retired and Sold.
ADM-2	View a pickup and return calendar.	Must	Today's tasks appear first.
ADM-3	See overdue rackets and outstanding refunds.	Must	Both appear on the home screen.
ADM-4	View pilot metrics.	Should	Shows the funnel in Section 16.
ADM-5	Keep an audit log of admin actions.	Must	Log cannot be edited from the app.
NTF-1	Prepare reminders before pickup and return.	Must	MVP uses a prefilled WhatsApp link sent by the operator.
NTF-2	Send automatic WhatsApp reminders.	Could	Added after the pilot.

7. Policies and business rules

All amounts in this section are proposed and should be tested. Publish them in plain language before the first paid trial.

7.1 Eligibility

- Customers must be 18 or older, with a verified Indonesian phone number.
- Staff check an ID in person before handover. The ID is returned at once and is not copied.
- New customers cannot book Tier 3 rackets until one trial is completed on time.

7.2 Fees and deposits

Racket value tier	Trial fee, 3 days	Refundable deposit	Late fee per day
Tier 1: under Rp1,500,000	Rp50,000	Rp250,000	Rp25,000
Tier 2: Rp1,500,000 to Rp3,000,000	Rp50,000	Rp500,000	Rp25,000
Tier 3: above Rp3,000,000	Rp50,000	Rp750,000	Rp25,000

- The trial fee is credited in full if the customer buys through Coba Raket within 14 days.
- Deposits are refunded within 24 hours of inspection, less any agreed charge.
- Cancellation is free at least 24 hours before pickup. After that, the trial fee is kept and the deposit is refunded.
- A racket more than 7 days late counts as unreturned. The deposit is kept and the balance of the agreed value is claimed.

7.3 Condition and damage grades

The grades adapt Blibli's approach of separating cracks, chips and fine scratches [7]. Normal play should never be charged.

Grade	What it means	Examples	Customer charge
Normal wear	Expected from play	Fine scratches, scuffs, overgrip wear, grommet wear, stretched strings	None
String break	Happens during normal hitting	Broken main or cross string	None. Treated as a running cost

Grade	What it means	Examples	Customer charge
Deep chip	Damage felt by touch, frame still safe	Deep paint and material chip from a hard court strike	Rp100,000 fixed
Structural	Frame is unsafe or cannot be repaired	Crack, bent frame or broken throat	Agreed racket value shown in the booking, less any resale value

7.4 Dispute process

1. Operator records the grade with evidence.
2. Customer receives the evidence and may respond within 48 hours.
3. The undisputed part of the deposit is refunded immediately.
4. A reviewer not involved in the booking decides within 3 days.
5. If evidence is unclear, the decision favours the customer.

Hidden cracks may exist before a trial. The inspection covers visible and touch-detectable damage only, and rental terms must say so. Under general rental rules, a renter may be liable for damage during the rental unless they show it was not their fault [22], but Coba Raket intentionally uses a fairer evidence rule to protect trust.

8. AI matcher specification

8.1 Design principle

Code decides what is allowed. The model only helps interpret free text and explain the result. It cannot invent a racket, change a price, or recommend something outside the stock database.

8.2 Flow

1. Collect level, playing frequency, purchase budget, grip size or hand guide, preferred feel, and any comfort concern.
2. Query rackets that are available on the requested dates.
3. Apply hard rules. For example, beginners start with lighter frames, commonly around 260 to 285 grams [17].
4. Score the remaining rackets using a small rules table.
5. Ask the model to write short explanations from a structured list of the selected rackets and their stored specifications.
6. Validate the JSON response. If any racket ID or number does not match, retry once, then use a rules-only template.

8.3 Output contract

Field	Type	Rule
racket_id	UUID	Must exist in the input list
fit_label	Enum	Easiest to start, Best balance, or Room to grow
reason	String	Up to 30 words in Bahasa Indonesia, using supplied facts only
tradeoff	String	One honest limitation
try_tip	String	One thing to notice during the trial

8.4 Guardrails

- Rate limit by device and phone number to control cost.
- Treat free text as data, not instructions.
- Do not ask for names, ID numbers or addresses in the matcher.
- For pain or injury, say the service cannot give medical advice, suggest a coach or health professional, and do not save the detail.
- Show that recommendations are suggestions and that the trial exists to confirm the fit.

8.5 Evaluation

Test	Method	Launch target
Coach agreement	Coach reviews 30 sample profiles	Coach accepts at least 80% of top-three lists
Stock accuracy	Automated test with changing inventory	0 unavailable rackets recommended
Fact accuracy	Compare all numbers with the database	0 invented or wrong values
Injection resistance	20 hostile prompts	0 rule changes or data leaks
Failure mode	Disable LLM provider	Rules-only shortlist still works

9. Racket passport specification

9.1 Why it exists

A used buyer must usually trust the seller. The passport makes it hard to rewrite a racket's recorded history after the fact. Blockchain is used for this one purpose, not for each handover decision or payment.

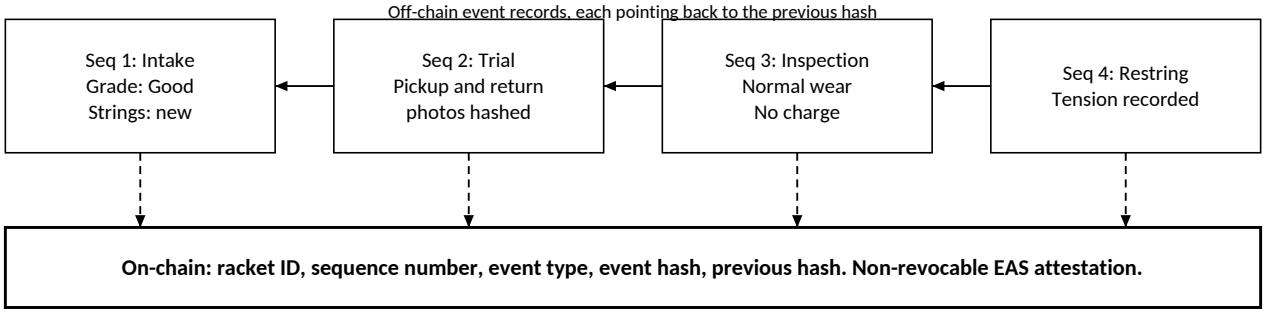


Figure 2. Each off-chain event links to the previous one and is anchored on-chain.

9.2 Design

Element	Decision
Network	Base Sepolia during the MVP. Move to mainnet only if used buyers value the feature
Attestation	Ethereum Attestation Service. Reported predeploys on Base are 0x4200000000000000000000000000000000000021 for EAS and 0x4200000000000000000000000000000000000020 for the schema registry [18]. Confirm before deployment
Schema	bytes32 racketId, uint32 seq, uint8 eventType, bytes32 eventHash, bytes32 prevHash
Revocation	Non-revocable. Corrections are new events that reference the earlier event
Event hash	Hash of canonical JSON plus a random salt. Racket photos are included by their file hash
Signing	One operator key held in server secrets. Pays testnet gas. Customers need no wallet
Private data	No names, phone numbers, ID data or booking IDs in event data or on-chain
Verification	The browser fetches events and attestations, recomputes each hash, then checks sequence and links

9.3 What it proves

Situation	Result	Why
Someone edits an old event in the database	Detected	The recomputed hash no longer matches
Someone deletes a middle event	Detected	Sequence and previous hash no longer connect
Someone backdates a new event	Detected	The attestation time and sequence show when it was recorded
Operator never records a crack	Not detected directly	A customer receipt and a required inspection step reduce this, but cannot remove it
Operator records false information	Not detected	A blockchain preserves a statement, not its truth
QR sticker moved to another racket	Reduced	Use a tamper-evident sticker, link any frame serial number, and show photos
Racket is a counterfeit	Not covered	The passport does not certify authenticity

10. User experience

10.1 Principles

- Bahasa Indonesia first, with plain words and no crypto vocabulary.
- Mobile web first, with no app install.
- One clear action per screen.
- Costs are always visible before commitment.
- Black text, large tap targets and readable contrast for outdoor use.

10.2 Key screens

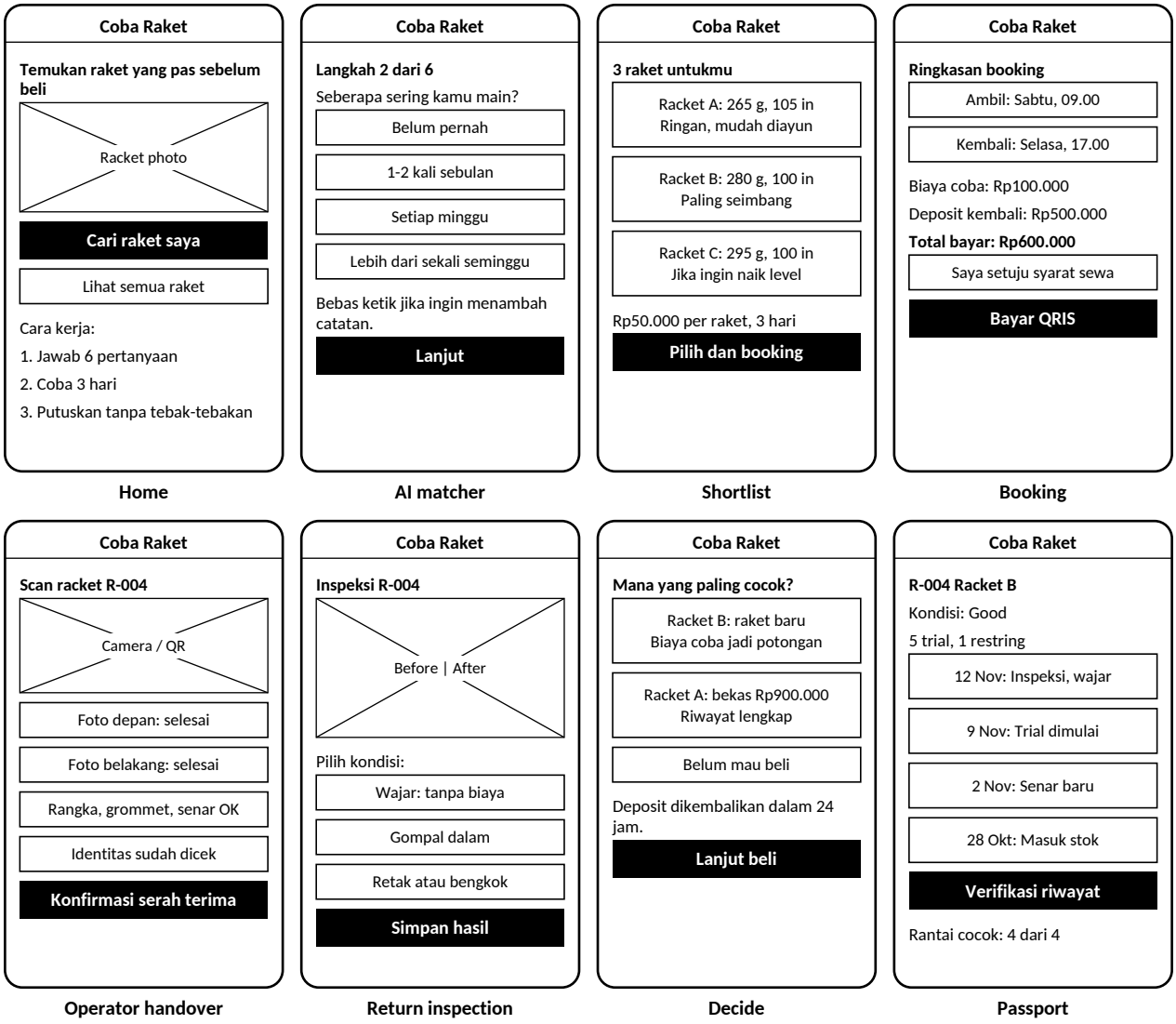


Figure 3. Low-fidelity wireframes. Screen text is sample copy and the crossed boxes are image placeholders.

10.3 Screen list

Area	Screens
Customer	Home, catalogue, racket detail, matcher, shortlist, booking summary, phone verification, payment, booking status, confirm handover, decide, dispute
Operator	Today, scan, handover, return inspection, overdue list, refunds, inventory, racket edit, metrics
Public	Passport, verification result, terms, privacy, how it works

11. Technical design

11.1 Architecture

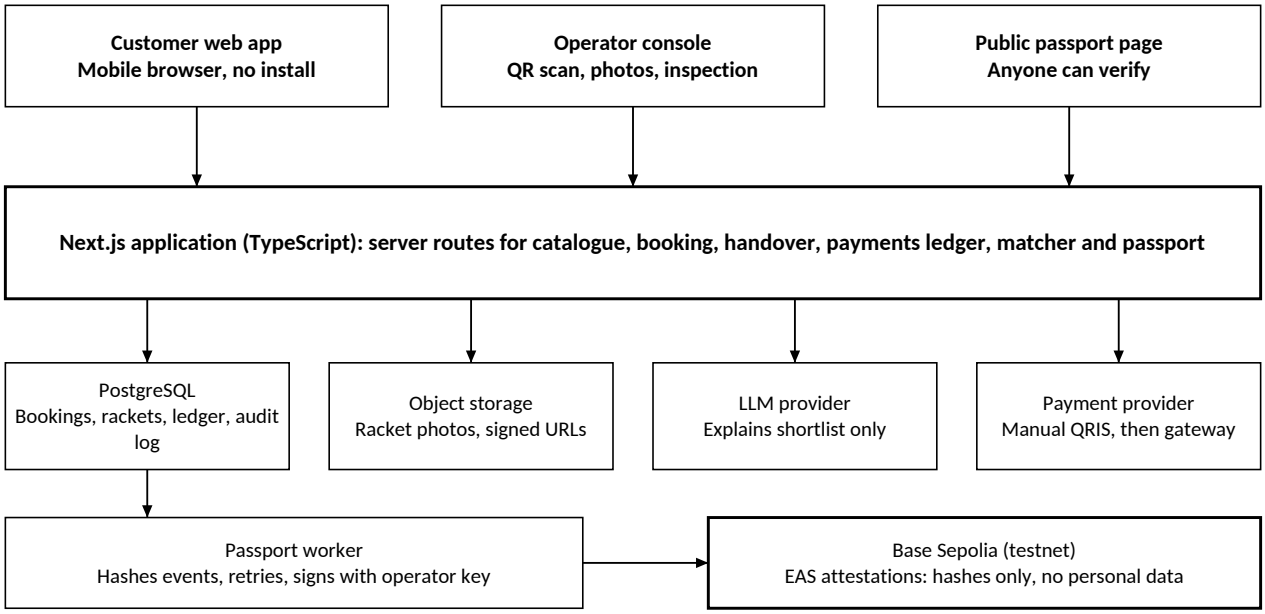


Figure 4. MVP system architecture.

11.2 Proposed stack

Layer	Choice	Reason
Frontend and backend	Next.js with TypeScript	One codebase, fast to deliver, strong portfolio value
Database and auth	PostgreSQL through Supabase, phone OTP	Relational data, row-level security, fast setup
File storage	Private object storage with signed URLs	Keeps racket photos out of public access by default
AI	Provider-neutral LLM adapter with JSON schema validation	Allows a change of model and supports the fallback path
Web3	viem, EAS SDK, Base Sepolia	No custom contract is needed in the MVP, which lowers risk
Hosting	Vercel and a scheduled worker	Simple deployment and retry jobs
Testing	Vitest, Playwright	Covers rules and critical journeys

11.3 Booking states

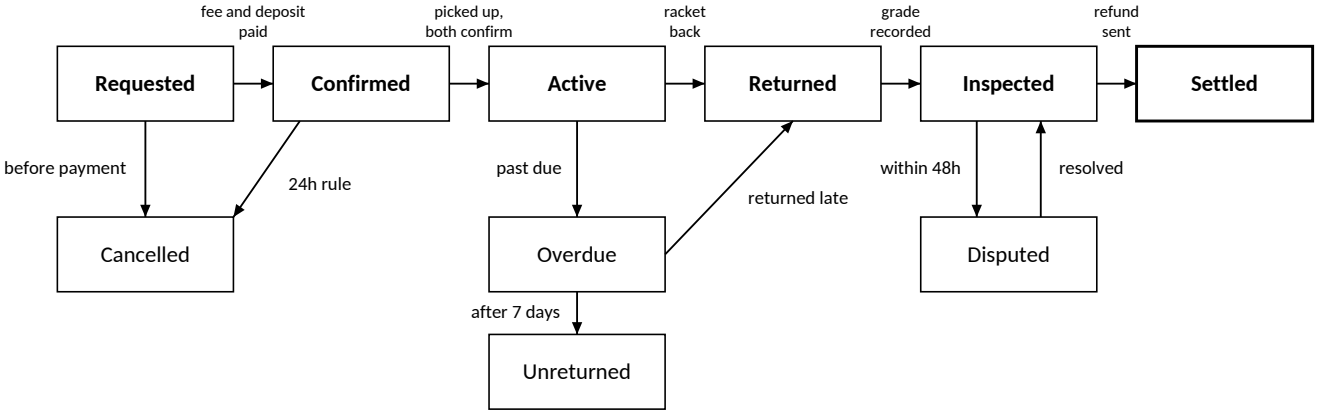


Figure 5. Booking state machine.

11.4 Data model

Table	Key fields
users	id, phone, display_name, role, verified_at, trials_completed
rackets	id, public_code, brand, model, weight_g, head_in2, grip, tier, condition, value_rp, status
bookings	id, user_id, pickup_at, return_due_at, state, terms_version, terms_accepted_at
booking_items	id, booking_id, racket_id, fee_rp, deposit_rp, rating, fit_note
ledger_entries	id, booking_id, type, amount_rp, direction, method, created_by, created_at
handovers	id, booking_item_id, type, operator_id, id_checked, checklist_json, customer_confirmed_at
photos	id, handover_id, side, storage_key, sha256, captured_at
damage_reports	id, booking_item_id, grade, charge_rp, reason, status, reviewer_id, decided_at
passport_events	id, racket_id, seq, type, payload_json, salt, event_hash, prev_hash, attestation_uid, status
match_sessions	id, anonymous_id, answers_json, shortlist_json, model_version, created_at
sales	id, racket_id, user_id, type, amount_rp, partner_id, credit_rp

11.5 Main API routes

Route	Purpose	Access
GET /api/rackets	List available stock	Public
POST /api/match	Create a shortlist	Public, rate limited
POST /api/bookings	Create a booking	Verified customer
POST /api/bookings/:id/payment	Record or confirm payment	Operator
POST /api/bookings/:id/handover	Record pickup or return	Operator
POST /api/handovers/:id/confirm	Customer confirmation	Booking customer
POST /api/items/:id/damage	Record grade and evidence	Operator
POST /api/damage/:id/dispute	Submit a dispute	Booking customer
GET /api/passport/:code	Fetch public history	Public

11.6 Non-functional requirements

Area	Requirement
Performance	Key pages usable within 3 seconds on mid-range phones over 4G. Matcher shortlist within 8 seconds
Reliability	Booking and handover continue if the LLM or blockchain is down
Security	HTTPS, role checks on every route, row-level security, rate limits, secrets outside code, two-factor sign-in for operators
Concurrency	Availability is enforced in the database, not only in the interface
Observability	Error tracking, failed passport event alerts, daily database backup
Accessibility	Tap targets at least 44 pixels, labelled controls, usable at 200% text zoom
Cost	Stay on free or low-cost tiers during the pilot. Keep LLM calls to a set daily limit

12. Privacy, legal and safety

This section is a product checklist, not legal advice. Review it with a qualified adviser before running a paid service at scale.

12.1 Personal data

Data	Purpose	Rule
Name and phone	Booking and contact	Delete or anonymise 12 months after the last booking, unless needed for a dispute (proposed)
ID document	In-person identity check	Do not copy, photograph or store. Save a true or false flag only
Comfort or injury mention	Immediate safety message	Do not save
Matcher answers	Improve recommendations	Anonymous and removed after 30 days (proposed)
Racket photos	Condition evidence	Show the racket only. Remove any photo that includes faces or documents
Payment records	Accounting and refunds	Retain as required. Do not store full bank credentials

- Publish a privacy notice and collect clear consent. Indonesia's Personal Data Protection Law (UU No. 27 of 2022) applies to this kind of processing.
- Keep personal data off-chain. Research warns that hashes can still count as personal data in some cases, so event hashes are salted and contain no personal data [19][20].
- Allow customers to request access, correction and deletion of their account data.

12.2 Agreements and evidence

The MVP records acceptance through phone verification and a checkbox. That is useful evidence, but an uncertified electronic signature is weaker than a certified one and may need extra proof in a dispute [21]. Add a licensed electronic signature provider only if larger deposits or disputes justify it.

12.3 Money handling

- Keep deposits separate from personal funds and record them in the ledger.
- Check before scaling whether deposit handling needs a licensed gateway or another arrangement.
- Register the business and confirm tax duties before regular operations begin.

12.4 Consumer and product safety

- Remove any cracked or bent racket from trials and from used sale listings unless clearly described and safe.
- Replace or clean grips between customers. Tennis Nuts uses protective head tape and overgrips on demo rackets [6].
- Do not claim authenticity unless the source of the racket is documented.
- Be honest that a trial reduces risk, but does not guarantee the perfect racket.

13. Business model

13.1 Revenue streams

Stream	How it works	Status
Trial fees	Rp50,000 per racket-trial (proposed)	Test first
Used racket sales	Sell retired stock with its history	Margin unknown
Referral commission	Partner stores pay for new racket sales	Rate must be negotiated
Stringing and accessories	Overgrips, strings and restring service	After pilot
Coach partnership	Coaches refer students and receive a share	A cost that drives demand

13.2 Unit economics

Assumptions: 6 rackets at Rp1,200,000 each, Rp50,000 per racket-trial, Rp10,000 per racket-trial for overgrip and cleaning, and Rp150,000 a month for hosting and restringing. A 3-day trial plus 1 turnaround day allows at most 7.5 racket-trials per racket each month.

Scenario	Trials per racket	Racket-trials per month	Use of capacity	Revenue	Costs	Monthly margin	Payback
Low	2	12	27%	Rp600,000	Rp270,000	Rp330,000	21.8 months
Base	4	24	53%	Rp1,200,000	Rp390,000	Rp810,000	8.9 months
High	6	36	80%	Rp1,800,000	Rp510,000	Rp1,290,000	5.6 months

The base case pays back in about 9 months. The high case needs 80% use of capacity, which is unlikely for a new service. Running costs are covered at about 4 racket-trials a month. The model excludes founder time, tax, payment fees, lost rackets, purchase margin and fee credits, so treat it as a rough guide.

If the 6 rackets later sell at an assumed 60% of cost, Rp4,320,000 of the Rp7,200,000 is recovered, leaving Rp2,880,000 of stock cost at risk.

13.3 Starting budget

Item	Lean option	Own stock option
Rackets	Borrowed: Rp0	Rp7,200,000
Tamper-evident QR stickers and printing	Rp100,000	Rp100,000
Overgrips and cleaning supplies	Rp300,000	Rp300,000
Domain and basic setup	Rp200,000	Rp200,000
Total estimate	**Rp600,000**	**Rp7,800,000**

14. Risks and mitigations

Risk	Likelihood	Impact	Mitigation
Low willingness to pay	High	High	Run the demand test first. Test Rp35,000 and Rp50,000 fees and a coach-led demo day
Free borrowing replaces the service	High	Medium	Focus on comparison, advice and premium models that friends may not own
A customer steals a racket	Medium	High	In-person ID check, deposit tiers, no Tier 3 for new customers, limited stock value
Deposit discourages booking	Medium	High	Partner with coaches, test lower deposits, track drop-off
Disputes harm trust	Medium	Medium	Clear grades, photos, customer-friendly rule when evidence is unclear
Hidden crack blamed on the wrong person	Medium	Medium	Charge only visible or touch-detectable damage, and inspect at both handovers
Few rackets create poor fit	Medium	Medium	Start with common grip sizes and a beginner-to-intermediate range
AI gives weak advice	Medium	Medium	Rules first, coach-reviewed test set, trial as final check
Passport is seen as a gimmick	Medium	Low	Hide blockchain language, explain the benefit as a verified history, measure use
Copycats appear	Medium	Medium	Build coach relationships, local trust and customer outcome data
Founder time is limited	High	Medium	Weekend-only pickup slots, clear cut list, and partner handover points
Regulatory or tax issue	Low	High	Seek advice before scale. Keep clear records

15. Go-to-market and validation

15.1 Phase 0: One-week demand test

Day	Action	Output
1 to 2	Interview 5 court owners, 3 coaches and 2 racket shops using Appendix A	Evidence and possible partner
3	Borrow 3 to 4 rackets, set a fee, write simple rental terms	Offer is ready
4	Create a form, QRIS payment and paper checklist	Manual flow is ready
5 to 7	Run a weekend demo day with a coach or court partner	Paid trials and buying signals

Gate 1 metric	Pass
Paid racket-trials	At least 10
Customers seriously ready to buy	At least 3
Partners	At least 1 coach or court
Returned on time and undamaged	All, or any issue is explainable and recoverable

15.2 Phase 1: Eight-week pilot

- Keep one pickup point near the most active court, with weekend slots first.
- Recruit 2 to 3 coaches through a simple referral arrangement.

- Reach players through campus tennis communities, local tournaments and short comparison videos.
- Run one demo day each month to create repeatable demand.

15.3 Later options

- Add padel. Rackets have no strings and many venues already rent them, but the market may be more crowded.
- Partner with stores that want to reduce return risk and drive premium sales.
- Consider self-service lockers only after demand is proven [23].
- Offer the system to other cities or clubs as software.

16. Success metrics

North Star metric: completed paid racket-trials that lead to a clear customer decision, either buy or not buy.

Stage	Metric	Pilot target (hypothesis)
Acquire	Visitors who start the matcher	At least 30%
Match	Started matchers that finish	At least 70%
Book	Completed matchers that lead to a paid trial	At least 15%
Operate	Median handover time	Under 60 seconds per racket
Trust	On-time returns	At least 90%
Trust	Trials ending in a dispute	Under 5%
Trust	Deposits refunded within 24 hours	At least 95%
Value	Customers rating the trial useful	At least 80%
Revenue	Racket-trials per month by month 2	At least 24
Revenue	Trial customers who buy within 14 days	At least 15%
Passport	Used buyers who open the passport	At least 50%

Gate 2: After the eight-week pilot

- **Scale:** at least 24 racket-trials in a month, 90% on-time returns, no more than one racket lost, and clear buying or referral income.
- **Change:** strong customer feedback but weak income. Test store partnerships or coach packages.
- **Stop:** fewer than 12 racket-trials a month after two changes in offer, or repeated losses.

17. Delivery plan

17.1 Two-week MVP build

Day	Work	Done when
1	Project setup, database schema, seed rackets, deployment	Live empty app with rackets in the database
2	Catalogue, detail page, availability rules	Customer can browse real stock
3	Rules-based matcher	Shortlist works without AI
4	LLM explanations, validation and fallback	AI text passes accuracy checks
5	Booking, phone verification and terms	Customer can create a booking
6	Manual payment and ledger	Operator can confirm payment
7	QR scan, photos and pickup handover	Pickup works on two phones
8	Return, damage grades, disputes and refunds	Booking can reach Settled
9	Passport events, EAS schema, worker and public verifier	Editing a record is detected
10	Decide flow, metrics, tests, security pass, launch checks	A real customer can complete the journey

17.2 If time runs short

Cut in this order: extensions, dashboard charts, customer receipts, automated AI free text, then the public verifier interface. Do not cut clear pricing, double-booking protection, handover proof, the ledger or privacy rules.

17.3 Launch checklist

- Rental terms and privacy notice are published.
- All critical tests pass on a real phone.
- Backups and error alerts are active.
- Operator practises five mock handovers.
- Two friends complete paid test trials before public launch.

18. Test plan

Area	Test case	Expected result
Booking	Two people book the same racket at once	One succeeds and one gets an availability message
Booking	Book during the turnaround day	Rejected
Payment	Confirm the same payment twice	One ledger entry
Refund	Refund more than the available deposit	Rejected
Handover	Scan a racket not in the booking	Blocked
Handover	Customer has not confirmed	Booking stays Confirmed
Damage	Add a charge without evidence	Blocked
Privacy	Inspect public passport payloads	No personal data
Security	Customer calls an operator route	Access denied
AI	Prompt asks for a free racket	Rules unchanged
Passport attack	Edit a stored event	Verifier shows mismatch
Passport attack	Delete a middle event	Verifier shows a gap
Passport failure	Network is unavailable	Handover succeeds, event stays Pending and retries

19. Open questions

Question	Why it matters	How to resolve
What trial fee feels fair?	Drives demand and revenue	Test Rp35,000 and Rp50,000
What deposit is acceptable?	Trades conversion against theft risk	Compare booking drop-off by tier
Can stores offer commission or stock?	Reduces capital and adds income	Talk to 2 local shops
Which court is the best first pickup point?	Drives convenience	Interview court owners and coaches
Do used buyers care about the passport?	Decides whether to keep or move it to mainnet	Test on first used sales
Which grip sizes are most needed?	Controls stock choice	Ask coaches and collect pilot data
What legal form suits the business?	Affects tax, contracts and deposits	Seek advice before scaling

Appendix A: Interview guide

Players

1. How did you choose your current racket? What did it cost?
2. Did you regret it? Why?
3. Have you tried another racket before buying? How?
4. What would you have paid to test three rackets for three days?
5. What deposit would stop you?
6. Would you buy a used racket if you could see its full history?

Coaches

1. What do students ask about buying rackets?
2. What mistakes do you see most often?
3. Do you lend rackets now? What goes wrong?
4. Would you recommend a trial service? What would you expect in return?

Courts and shops

1. Do customers ask to borrow or rent rackets?
2. What do you charge, and how often are rackets damaged or not returned?
3. Would you host a pickup point or demo day?
4. Would you offer a commission or demo stock?

Appendix B: Rental terms outline

1. Parties, racket details, agreed value, trial dates and return place.
2. Trial fee, deposit, refund timing and fee credit.
3. Permitted use and customer care duties.
4. Meaning of normal wear, deep chip and structural damage.
5. Limits of inspection, including hidden damage.
6. Late returns, non-return and cancellation.
7. Dispute process and evidence.
8. Personal data notice.
9. Limits of advice, including no guarantee of fit.
10. Contact details and version date.

Appendix C: Inspection checklist

Part	Check	Record
Frame head	Scratches, chips, cracks or bends	Photo and condition
Throat	Cracks by sight and touch	Pass or fail
Grommets and bumper	Broken or missing parts	Pass or note
Strings	Breaks, heavy fraying and approximate tension	Pass or replace
Grip	Tears, odour or looseness	Replace overgrip
Butt cap and handle	Loose or broken cap	Pass or note
Identity	QR sticker intact and any serial number matches	Pass or stop

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Sources were reviewed on 4 November 2026. Listing prices can change. Some are single listings, and they should be treated as examples rather than market averages.